

ODE versus PDE

Population equation $\frac{dP}{dt} = P$

General solution $P(t) = C e^t$
 $\uparrow$
 arbitrary constant

PDE that looks like Pop eqn $u_t^+ = u$ (assume u depends on x, t)

General solution? $u(x, t) = C e^t$

$$u(x, t) = F(x) e^t$$

$\uparrow$
 arbitrary function

Introduction to PDEs

1) What are PDEs? 2) Why should you care? 3) What types of PDEs are there?

1) PDE = partial differential equation, an equation involving a function and/or its partial derivatives.

e.g. $\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}$ $\frac{df}{dx} = f^2$ (ODE)

$\underbrace{\hspace{10em}}_{f(x, t)} \quad \underbrace{\hspace{10em}}_{f(x)}$

2) Heat Conduction, Fluid Motion (Navier-Stokes), Schrödinger's Equation, General Relativity (EFE).

3) a) Order = Number corresponding to order of highest partial derivative in the equation.

Order 2 $\frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t}$ - (1) Order 3 $\frac{\partial^3 f}{\partial x^2 \partial y} = \frac{\partial f}{\partial t}$ - (2)

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 2 (x,t) ind. var 3 (x,y,t) ind. var

b) No. of independent variables = Independent variable : what we differentiate with respect to.

c) Linearity = A linear PDE is one in which the dependent variable (the one being differentiated) appears only in a linear fashion.

$\frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t}$ - (1) $\frac{\partial^3 f}{\partial x^2 \partial y} = \frac{\partial f}{\partial t}$ - (2) → both are linear

(Still linear) $\frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t} \cdot \tan t$ → $\frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t} \cdot \tan t$ nonlinear

4) Homogeneity = A homogenous PDE is one in which every term only involves the dependent variable and/or its derivatives.

$\frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t}$ - (1) $\frac{\partial^3 f}{\partial x^2 \partial y} = \frac{\partial f}{\partial t}$ - (2) } homogenous

$\frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t} + x^2 + \tan t$ } not homogenous

5) Coefficient Type = Constant or Variable.

$1 \frac{\partial^2 f}{\partial x^2} = 1 \frac{\partial f}{\partial t}$ - (1) $1 \frac{\partial^3 f}{\partial x^2 \partial y} = 1 \frac{\partial f}{\partial t}$ - (2) } Constant coeffs.

$\tan x \frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t}$ } variable coeffs.

Basic Concepts of PDEs

A partial differential equation (PDE) is an equation involving one or more partial derivatives of an (unknown) function, call it u , that depends on two or more variables, often time t and one or several variables in space. The order of the highest derivative is called the order of the PDE. Just as was the case for ODEs, second-order PDEs will be the most important ones in applications.

Just as for ordinary differential equations (ODEs) we say that a PDE is linear if it is of the first degree in the unknown function u and its partial derivatives. Otherwise we call it nonlinear. Thus, all the equations in Example 1 are linear. We call a linear PDE homogeneous if each of its terms contains either u or one of its partial derivatives. Otherwise we call the equation nonhomogeneous. Thus (4) in Example 1 (with f not

of the first degree in the unknown function u and its partial derivatives. Otherwise we call it **nonlinear**. Thus, all the equations in Example 1 are linear. We call a **linear PDE homogeneous** if each of its terms contains either u or one of its partial derivatives. Otherwise we call the equation **nonhomogeneous**. Thus, (4) in Example 1 (with f not identically zero) is nonhomogeneous, whereas the other equations are homogeneous.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{One-dimensional wave equation}$$

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{One-dimensional heat equation}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{Two-dimensional Laplace equation}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y) \quad \text{Two-dimensional Poisson equation}$$

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad \text{Two-dimensional wave equation}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \quad \text{Three-dimensional Laplace equation}$$

2nd Order, Linear &

$$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} - 3xu_x + 4yu_y + 5u = 0$$

2nd O, Nonh, Lin

$$u_{xx} + 2u_{xy} + u_{yy} + xu_x - uy_y + u = e^x \sin y$$

Order
Linear or nonlinear
Homogeneous or not?

$$u_{xxx} + xu_{xy} + u_x u_y + u^2 = 0$$

Non linear

$$(1 + x^2)u_{xx} - 2xyu_{xy} + y^2u_{yy} + 3u_x - 2u = x^2 + y^2$$

$$u_{tt} - c^2 u_{xx} + u u_x = \sin(x + t)$$

u, linear, Nonh

$$x^2 u_{xxx} + 2xy u_{xxy} + y^2 u_{xyy} - 5xu_x + 7u = 0$$

A **solution** of a PDE in some region R of the space of the independent variables is a function that has all the partial derivatives appearing in the PDE in some domain D (definition in Sec. 9.6) containing R , and satisfies the PDE everywhere in R .

Often one merely requires that the function is continuous on the boundary of R , has those derivatives in the interior of R , and satisfies the PDE in the interior of R . Letting R lie in D simplifies the situation regarding derivatives on the boundary of R , which is then the same on the boundary as it is in the interior of R .

In general, the totality of solutions of a PDE is very large. For example, the functions

$$(7) \quad u = x^2 - y^2, \quad u = e^x \cos y, \quad u = \sin x \cosh y, \quad u = \ln(x^2 + y^2)$$

which are entirely different from each other, are solutions of (3), as you may verify. We shall see later that the unique solution of a PDE corresponding to a given physical problem will be obtained by the use of **additional conditions** arising from the problem. For instance, this may be the condition that the solution u assume given values on the boundary of the region R ("**boundary conditions**"). Or, when time t is one of the variables, u (or $u_t = \partial u / \partial t$ or both) may be prescribed at $t = 0$ ("**initial conditions**").

We know that if an ODE is linear and homogeneous, then from known solutions we can obtain further solutions by superposition. For PDEs the situation is quite similar:

THEOREM 1

Fundamental Theorem on Superposition

If u_1 and u_2 are solutions of a homogeneous linear PDE in some region R , then

$$u = c_1 u_1 + c_2 u_2$$

with any constants c_1 and c_2 is also a solution of that PDE in the region R .

EXAMPLE 2 Solving $u_{xx} - u = 0$ Like an ODE

Find solutions u of the PDE $u_{xx} - u = 0$ depending on x and y .

Solution. Since no y -derivatives occur, we can solve this PDE like $u'' - u = 0$. In Sec. 2.2 we would have obtained $u = Ae^x + Be^{-x}$ with constant A and B . Here A and B may be functions of y , so that the answer is

$$u(x, y) = A(y)e^x + B(y)e^{-x}$$

with arbitrary functions A and B . We thus have a great variety of solutions. Check the result by differentiation. ■

EXAMPLE 3 Solving $u_{xy} = -u_x$ Like an ODE

Find solutions $u = u(x, y)$ of this PDE.

Solution. Setting $u_x = v$, the PDE becomes $v_y = -v$. Integrating with respect to y , we get

$$v = f(x)e^{-y} \quad \text{where} \quad f(x) = \int g(x) dx$$

here, $f(x)$ and $g(y)$ are arbitrary. ■

15. Boundary value problem. Verify that the function

$$u(x, y) = a \ln(x^2 + y^2) + b$$

satisfies Laplace's equation (3) and determine a and b so that u satisfies the boundary conditions $u = 110$ on the circle $x^2 + y^2 = 1$ and $u = 0$ on the circle $x^2 + y^2 = 100$.

16–23 PDEs SOLVABLE AS ODEs

This happens if a PDE involves derivatives with respect to one variable only (or can be transformed to such a form), so that the other variable(s) can be treated as parameter(s). Solve for $u = u(x, y)$:

16. $u_{yy} = 0$

17. $u_{xx} + 16\pi^2 u = 0$

19. $u_y + y^2 u = 0$

20. $2u_{xx} + 9u_x + 4u = -3 \cos x - 29 \sin x$

21. $u_{yy} + 6u_y + 13u = 4e^{3y}$

22. $u_{xy} = u_x$

23. $x^2 u_{xx} + 2x u_x - 2u = 0$

24. **Surface of revolution.** Show that the solutions $z = z(x, y)$ of $yz_x = xz_y$ represent surfaces of revolution. Give examples. *Hint.* Use polar coordinates r, θ and show that the equation becomes $z_\theta = 0$.

25. **System of PDEs.** Solve $u_{xx} = 0, u_{yy} = 0$

Heat equation $u_t = u_{xx}$

$u = \frac{1}{2}x^2 + t$ is a solution.

$u = e^{ax+bt}$, what are a, b ?